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A Weight Multi-Net Architecture for Bearing Fault Classification under Complex Speed Conditions

Xiaoxi Ding (*Member IEEE*), Lun Lin, Dong He, Liming Wang, Wenbin Huang and Yimin Shao

***Abstract*—As a key component in rotating mechanical, fault identification of bearing is a hot topic for machine health condition monitoring. Considering the corresponding vibration features are seriously influenced by working conditions, the existed classification methods have high false positive rate under complex working conditions. Specially, because there is a difference among feature distribution of the measured signals under different speed conditions, the fault identification under uncertain speed conditions is still a big challenge for the fault diagnosis schemes based on expert experience and intelligent fault identification method. Therefore, this study proposes a weight multi-net architecture, which consists of multiple net units. Different from other traditional networks, the exclusive characteristics of fault information will be mined via a multi-class identification unit, and the vital information related to speed condition will be remained by a weight unit. Then, these fault information with different directivity will be integrated in a fusion unit. Finally, the wavelet packet energy ratios of vibration signals are extracted and put into this special architecture of neural network with three function units, and the effect of fault classification under unknown speed conditions can be significantly improved. Comparisons of clustering distribution and classification accuracy with other typical methods show the feasibility and effectiveness of the proposed weight multi-net method in the application of fault identification under uncertain speed conditions.**

***Index Terms*—Fault classification, multi-class identification unit, weight multi-net, fusion unit, complex speed conditions**

I. Introduction

WITH the development of the manufacturing industry, much more important equipment is widely used in the manufacturing process of various industries. Their failures would cause serious damage and even safety incidents [1-3] in the case of expensive cost and critical using areas. Thus, health monitoring and fault diagnosis technique is a crucial issue in equipment state monitoring system. As the emergence and development of feature analysis and machine learning, the data-driven classification method provides a powerful ways for the identification of equipment health status, such as SVM [4-6], decisions trees [7-8], random forest algorithm [9-10], Fuzzy measurement [11], and Bayesian Network(BN) [12-13]. Compared with other techniques, by imitating the structure of human neural and representing the information related to classification implicates by multiple layers consist of a number of nerves, artificial neural network (ANN) has irreplaceable advantages in pattern classification [14-17]. In which, artificial neural networks (ANN) have been widely used in the process of mechanical fault identification due to its simple composition and solid theoretical foundation [18-19].

Generally, the traditional fault identification system consists of three parts: signal processing, feature extraction, and model construction [20]. With the development of pattern recognition technologies, many exist methods have tried to improve the first two parts in a much more concise and intelligent way, where the original signal or spectrum are directly employed for fault identification [21-22]. However, the architecture and the training process have an important influence on the data flow mode and information extraction effectiveness of the model. With an assumption that each layer is trying to mine the valid information, the performance of a network with simple structure would drop sharply when solving a complex identification issue. Therefore, the network architecture design is the vital mainstream to solve complex pattern recognition problem. According to the principles of the framework optimization and data transfer, researchers has built many new network models, such as convolutional neural network (CNN) and deep neural network (DNN). These new network models with special architectures have shown their strong potential in complex health detection and fault recognition [23-25].

As we known, much equipment often work in a complex and terrible conditions, and the speed information is always missing while the operating conditions are even unknown. The speed information is not only crucial for periodic characteristics analysis but also has a great affection to identification result of intelligent fault identification algorithms due to its influence to the distribution of fault signal. It can be foreseen that the traditional fault analysis methods have some shortcomings in the identification of bearing faults once the speed conditions are unknown. Nevertheless, many fault identification approaches have lost sight of the influence of

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speed conditions on characteristics distribution in the process of model construction. Thus, a classification model trained by those fusion samples under several known speeds may turn into fail once the input samples under other unknown conditions were given. This has severely limits the application of intelligent methods in the field of equipment fault identification. It should be noted that some researchers have noticed this phenomenon and conducted some valuable research. Han [26] selects some insensitive features for load via a new feature extraction method and combine these features with SVM, which aims to solve the problem of gear fault identification under unstable load conditions in some level. Ding [27] uses the energy ratio of wavelet coefficients as the input features of a multi-scale Conv-Net structure for fault identification of rolling bearings, and the model has certain fault recognition ability under the different conditions. Pramod [28] built a wind turbine gearbox maintenance management system based on the self-evolution network model by using the SCADA data. Xiong [29] added a dynamic weighted layer to the cross-layer connection of the densely to strengthen the deep information transmission of the network, and proposed a fault diagnosis method for the dynamically weighted densely connected convolutional network. On one hand, there are no valid features which are completely subject to the working condition. It can be seen that this crucial speed information cannot be well translated and represented. On the other hand, when the working condition is relatively vague and changeable, a higher network complexity is not an ideal way to get a high classification performance. Thus, the method of fault identification under uncertain conditions is a significant research issue.

Base on the principle of artificial neural network, this paper proposed a new weight multi-net architecture (WMN), which consists of fault identification unit, weight unit and fusion unit. During the training process of this special fault analysis method, the fault information under different speed conditions is first extracted by different sub-nets of fault identification unit. Then the laws of feature distribution influenced by speed are further built and expressed with a series of output weights by the weight unit. Finally, the identification outputs of fault sub-nets are fused with those output weights by fusion unit. Different from the traditional fault identification model, by synthesizing fault classification information and speed information via different units, the proposed weight multi-net not only well solves the problem of mechanical fault identification under unknown speed conditions, but also provides a potential method for the problem of classification with complex working conditions.

The remainder of this paper is organized as follows. Section II presents a basic theory of ANN and WPT. In section III proposes WMN architecture and introduce the process of information transmission. Section IV investigates the effectiveness and efficiency of the proposed WMN method for bearing fault classification under complex speed and discusses some worth-thinking appearance. The conclusions are drawn in section V.

II. Theoretical Background

A. Wavelet Packet Transform

Wavelet packer Transform (WPT) is an effective signal transform technique by decomposing a non-stationary signal into multiple sub-signals with an advantage of multiple resolutions, which can avoid the weakness of discrete wavelet transform (DWT) [30,31]. The raw signal is decomposed by a series of low-pass filters *h(k)* and high-pass filters *g(k)* as shown:

$$w_{i+1}^{2s}(t) = \sum_k h(k) w_i^s(2t-k) \tag{1}$$

$$w_{i+1}^{2s+1}(t) = \sum_k g(k) w_i^{s+1}(2t-k) \tag{2}$$

where the raw signal $x(t)$ is $w_0^1(k)$,(k=1,2...S, S is the decomposition level) and $w_i^s(k)$ is the WP decomposition coefficients of node s at level i. And $w_{i+1}^{2s}(k)$ and $w_{i+1}^{2s+1}(k)$ are respectively the wavelet coefficients of node $2s$ and $2s+1$ at level $i+1$, which correspond to the approximation coefficients and detail coefficients.

In this manner, it can be foreseen that the raw signal will be decomposed into a comprehensive frequency band. At level i, 2^i sub-signals can be obtained and the corresponding nodes can be identified as a binary decomposition tree which is illustrated as Fig. 1 (where a raw signal is decomposed through a 3-level WPT).

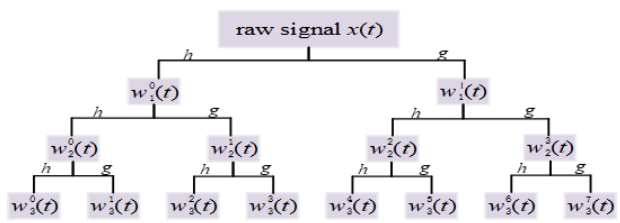

Fig. 1. Binary tree of WPT

B. Conventional Neural Network

The typical ANN is a full interconnection network which consists of input layer, hidden layer and output layer [32]. The detail of BP Net is shown as Fig. 2.

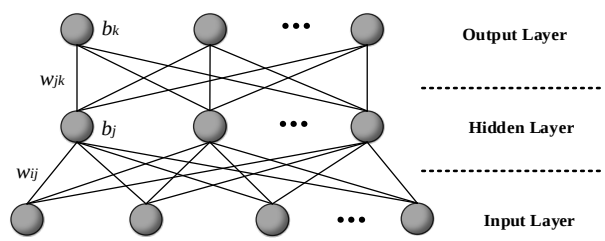

Fig. 2. Traditional structure of ANN Net

The training process of ANN includes two parts, forward propagation of features and back propagation (BP) of deviation [32-34]. In the process of forward propagation, the input features x_i (i=1,2,…,N) are first transmitted from the input layer to the output layer through the hidden layers. The output of the unit j in the hidden layer and the unit k in the output layer can be written as follows.

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$$o_j = \phi(\sum_{i=1}^{N} \omega_{ji} x_i + b_j) \qquad (j=1,2,.....M) \tag{3}$$

$$o_k = \psi\left(\sum_{j=1}^{M} \omega_{kj} \phi(\sum_{i=1}^{N} \omega_{ji} x_i + b_j) + b_k\right) \quad (k=1,2.....L) \tag{4}$$

It should be noted that φ and ψ represent the transfer function of hidden layer and output layer. Parameters N, M and L are respectively the number of the units in input layer, hidden layer and output layer. The value w_{ji} and w_{kj} are the weight of the unit i in input layer and unit j in hidden layer and the unit j in hidden layer and unit k in output layer. And value b_j and b_k are the threshold variable of the unit j in hidden layer and the unit k in hidden layer.

In the process of back propagation, the parameters are trained by the negative gradient method in optimization theory, where the cost function is calculated by the square error function [19]. The square error function is shown as:

$$E = \frac{1}{2}\sum_{i=1}^{M} \|d_i - y_i\|^2 \tag{5}$$

In which, y_i is the output of model, and d_i is the ideal output of training data.

III. Weight Multi-net Architecture

A. WMN Architecture

The architecture of the proposed weight multi-net is shown as Fig. 3, which consists of four parts, the input features, identification unit, weight unit and fusion unit. First, several sub-nets (V_1_Net, V_2_Net ... V_n_Net) are combined to form the fault identification unit, where each sub-net contains the fault types under the same speed condition. As we known that, each sub-net is trained by the samples of a typical working condition. This further indicates that multi-classifiers are built for fault types with the corresponding speed information remained. Thus, the fault attributive information for different typical working conditions is mined in different sub-nets. Once a fault sample was put into these multi-classifiers, the corresponding similar classifier would be activated while other rest classifiers would show an inconsistent response. Second, a weight unit is constructed aim to extraction the related information of speed for input features, and a series of weight can be built for the fault analysis results for each sub-net of typical speed. These fault identification information via each sub-net are further fused with the output weights in the layer of fusion unit. The final classification result is obtained.

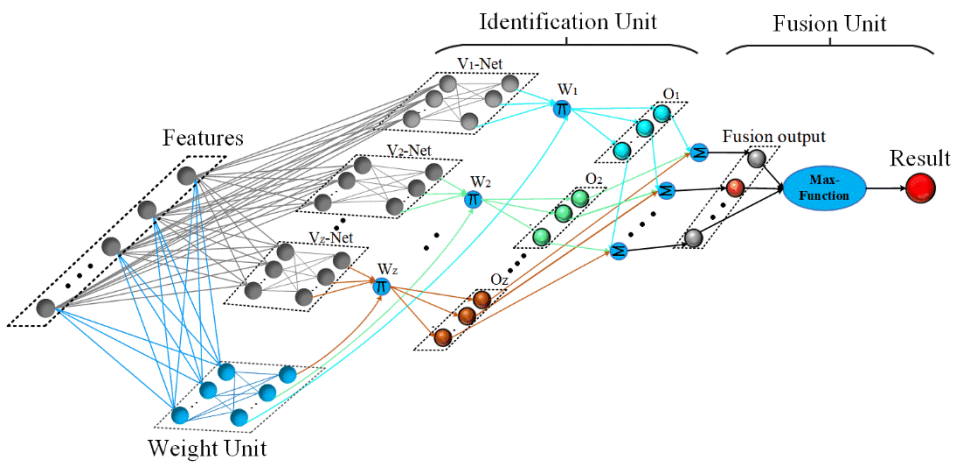

Fig. 3. Illustration of WMN architecture.

In order to achieve the target recognition accuracy in the training process of the model, the weight unit tends to assign large weight to the correctly classified sub-nets in the fault identification unit, and assign small value to other sub-nets with incorrect classification. This indicates that there is a dynamic weight adjustment process before the feature fusion unit. In the process of multi-nets training, the sub-net classifier is trained by the samples from the same working speed condition while the weight unit is trained by the samples from the same fault type. In these manners, the laws of feature distribution with working speed condition are extraction and activated. This trained network model will be further used for fault classification under unknown speed conditions. It can be known that once the working condition of a new input sample is similar to the condition information of a typical sub-net in the fault identification unit, the weight unit would assign a larger weight value to this sub-net while other sun-net is suppressed with a smaller weight value, so that the mostly correct recognition result is obtained in the fusion unit. If the working condition of input sample is not similar to all sub-nets in fault identification unit, the weight unit will tend to assign comprehensive weights to all sub-nets, and by fusing the classification results from different sub-nets together, the WMN can achieve a higher recognition precision for the samples under unknown working conditions. According to the principle of WMN, it can be seen that, with fault classification information activation and speed condition information exaction in different units, sensitive fault features that are robust to speed conditions are built in fusion unit. Therefore, this study constructs a speed-insensitive fault identification network structure with the consideration of the speed influence pattern on the signal characteristics. It should be noted that,

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according to the principle of ANN [16], *sigmod* function is used for the nets in the identification net and *max*-function is employed as output function in the fusion unit.

B. Fault Identification Unit with Multi-classifiers

Different from traditional identification method, this architecture is first to construct multi-classifiers due to the influence of speed condition in the pre-training process. Multiple sub-nets subjected to the same speed condition are first built to follow the typical three-layer BP network, which are respectively named as V_1-*Net*、V_2-*Net* ..... V_Z-*Net* (z=1，2 ... Z, Z is the given number of the speed conditions) as shown in Fig 4. Similar with the typical BP network, the architectures of these sub-nets are also pre-trained by gradient descent algorithm. It should be noted that the different identification sub-net is pre-trained by the different fault samples under the corresponding speed conditions. Once the testing accuracy achieved the preset objective, the information of fault classification under each speed nodes has been extracted and stored in different sub-nets. The identification result of sub-net z is signed as $O_z=[o_{z\,1},o_{z\,2}\ldots o_k^z,\ldots,o_L^z]$, where the result o_k^z is the probability vector of the input sample k belongs to fault type z. Here, L is the number of nodes in output layer, which equal to the number of fault types. Therefore, the final fault identification unit is constructed for the next unit.

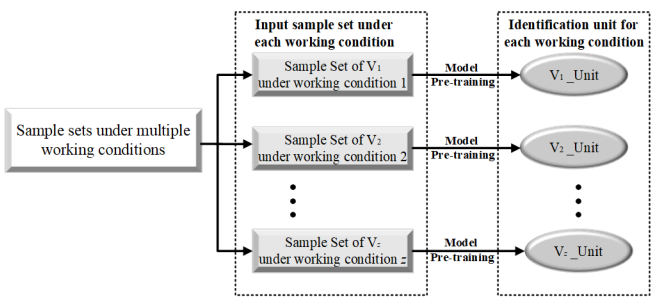

Fig. 4. The pre-training procedure of fault identification unit.

C. Weight Unit for Working Conditions Representation

To solve the limitation of the speed influence in the feature identification, the proposed architecture introduces a weight unit to extract the features related to speed information. Figure 5 illustrates the mechanism between weight unit and fusion unit, where the output of multi-classifiers is used as the intermediate input for fusion unit. This means that there exist a new four-layer weight network consisting of input layer, hidden layer, weight calculated layer, and decision fusion layer. The input of the weight unit is a feature vector of vibration signal while the output layer is a fusion weight vector, which corresponds to the outputs of the multiple sub-nets under different speed conditions in fault identification unit. The multiple recognition results are further linearly fused by the weight vector in fusion unit, and the corresponding classification label of the input sample is achieved via a max-function. It can be seen that the dynamic weight information will be well fused in this process.

The output of unit j in hidden layer and unit k of output layer in weight unit are respectively shown as following:

$$o_j^w = \phi(\sum_{i=1}^{N} \omega_{ji} x_i + b_j) \qquad j = 1,2,\ldots,M \tag{6}$$

$$\omega_z = \psi(\sum_{j=1}^{M} \omega_{zj} \phi(\sum_{i=1}^{N} \omega_{ji} x_i + b_j) + b_z) \qquad z = 1,2,\ldots,Z \tag{7}$$

where φ and ψ represent the transfer function of hidden layer and output layer in weight unit. Then the fusion result O can be further calculated by the following function

$$\begin{aligned} O &= \omega_1 O^1 + \omega_2 O^2 + \ldots\ldots + \omega_Z O^Z \\ &= [\omega_1, \omega_2 \ldots\ldots, \omega_Z] \begin{bmatrix} o_1^1 & o_2^1 & \ldots & o_L^1 \\ o_1^2 & o_2^2 & \ldots & o_L^2 \\ \ldots & \ldots & \ldots & \ldots \\ o_1^z & o_2^z & \ldots & o_L^z \end{bmatrix} \\ &= [o_1, o_2 \ldots o_L] \end{aligned} \tag{8}$$

where parameters N, M and L are respectively the number of the units in input layer, hidden layer and output layer for this weight unit. The value w_{ji} and w_{zj} in this weight unit stands for the weight of the unit i in input layer and unit j in hidden layer and the unit j in hidden layer and unit z in output layer. And value b_j and b_k are the threshold variable of the unit j in hidden layer and the unit k in hidden layer of weight unit.

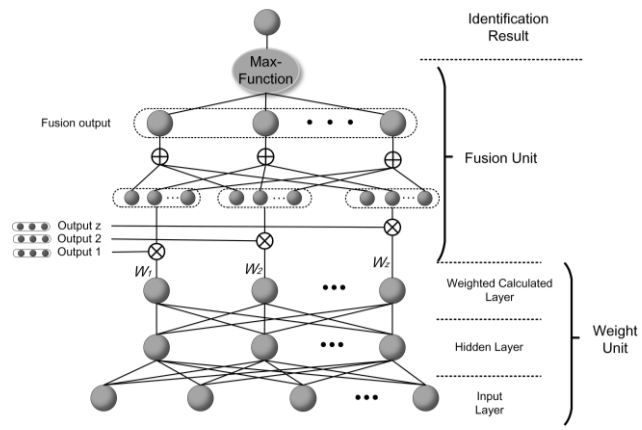

Fig. 5. The four-layer weight network architecture.

D. WMN Training for Fusion Information Extraction

In the proposed WMN architecture, the characteristics related to speed information are extracted by weight unit. Thus, the training process of weight unit is the key part in the process of network model designing. With the employment of the fusion unit for identification result searching, the fusion net has to be optimized in the process of weight unit training. Suppose the goal of a sample is $T=[t_1,t_2\ldots t_L]$. According to the output function of weight unit and fusion unit, the derivative and training error of output result O to the output weight w_z are shown as follows

$$\frac{\partial O}{\partial w_z} = O^z \tag{9}$$

$$E_{wz} = \frac{\partial O}{\partial \omega_z}(O - T) = O^z (O - T) = \sum_{l=1}^{L} o_l^Z (o_l - T_l) \tag{10}$$

The structural parameters in weight unit are also optimized as:

$$b_z = b_z + \Delta b_z \tag{11}$$

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$$\omega_{zj} = \omega_{zj} + \Delta\omega_{zj} \tag{12}$$

$$b_j = b_j + \Delta b_j \tag{13}$$

$$\omega_{ji} = \omega_{ji} + \Delta\omega_{ji} \tag{14}$$

Thus, the correction quantity of structural parameters can be calculated as

$$\Delta b_z = -\eta \sum_{l=1}^{L} o_l^z \left(o_l - t_l\right) \psi'\left(\sum_{j=1}^{M} \omega_{zj} \phi\left(\sum_{i=1}^{N} \omega_{ji} x_i + b_j\right) + b_z\right) \tag{15}$$

$$\Delta\omega_{zj} = -\eta \sum_{l=1}^{L} o_l^z \left(o_l - t_l\right) \psi'\left(\sum_{j=1}^{M} \omega_{zj} \phi\left(\sum_{i=1}^{N} \omega_{ji} x_i + b_j\right) + b_z\right) \phi\left(\sum_{i=1}^{N} \omega_{ji} x_i + b_j\right) \tag{16}$$

$$\Delta b_j = -\eta \sum_{z=1}^{Z} \sum_{l=1}^{L} o_l^z \left(o_l - t_l\right) \psi'\left(\sum_{j=1}^{M} \omega_{zj} \phi\left(\sum_{i=1}^{N} \omega_{ji} x_i + b_j\right) + b_z\right) \omega_{zj} \phi'\left(\sum_{i=1}^{N} \omega_{ji} x_i + b_j\right) \tag{17}$$

$$\Delta\omega_{ji} = -\eta \sum_{z=1}^{Z} \sum_{l=1}^{L} o_l^z \left(o_l - t_l\right) \psi'\left(\sum_{j=1}^{M} \omega_{zj} \phi\left(\sum_{i=1}^{N} \omega_{ji} x_i + b_j\right) + b_z\right) \omega_{zj} \phi'\left(\sum_{i=1}^{N} \omega_{ji} x_i + b_j\right).x_i \tag{18}$$

where η is the study rate. To avoid the problem of slow convergence and local optimum, an adaptive algorithm is used to adjust the study rate, where the study process is dynamically adjusted according to the training error in each step. The empirical coefficients and update method of study rate in step k+1 is shown as following [16]:

$$\eta(k+1) = \begin{cases} 1.1\eta(k) & ep(k) < 0.9ep(k-1) \\ 0.85\eta(k) & ep(k) > 1.05ep(k-1) \\ \eta(k) & other \end{cases} \tag{19}$$

where $\eta(k)$ and $ep(k)$ are respectively the study rate and training error after step k. Similar with Eq. (5), thee training error ep between the training result and training targets will be optimized by BP method. In the training process, the output weights are adjusted by the error minimization between the training targets and the output of the sub-nets in identification unit. For a smaller training error, the weight unit tends to fine tuning parameters to increase the weight of the corresponding sub-nets. Otherwise, the structural parameters will be tuned to decrease the weight. For a new sample, the multi-classifiers provided by those sub-nets aim to distinguish it into the right category with the same working condition. Therefore, the crucial information of fusion weight and feature distribution with same speed condition are mined and memorized in this weight unit.

In summary, the information about fault classification with typical speed nodes has been remained in different sub-nets in identification unit. Meanwhile the information of fusion weight distribution subjected to speed conditions has been also extracted by weight unit. Thus, once even a sample with unknown condition was further analyzed, the proposed WMN can achieve a high accuracy for fault identification under unknown speed conditions.

E. Bearing fault identification via multi-weight fusion

Once the WMN model was constructed, the bearing fault under unknown speed condition can be identified with multi-weight fusion. The identification process is illustrated as Fig. 6, which consists of three parts: feature construction, weight multi-net net model training, and fault identification. Firstly, the raw vibration signals are obtained, where multiple fault types of bearing under different speed conditions are used. They are decomposed into multiple sub-signals via WPT and the fault sample sets are built by the decomposition coefficients energy ratio from each sub-signal. Then, the model training set and testing set are divided, and a weight multi-net architecture with three sub-nets is built and trained by the training sets of several typical speed conditions. Finally, the trained model is tested by the testing sets and a classification model with high precision for bearing fault identification under uncertain speed condition is obtained.

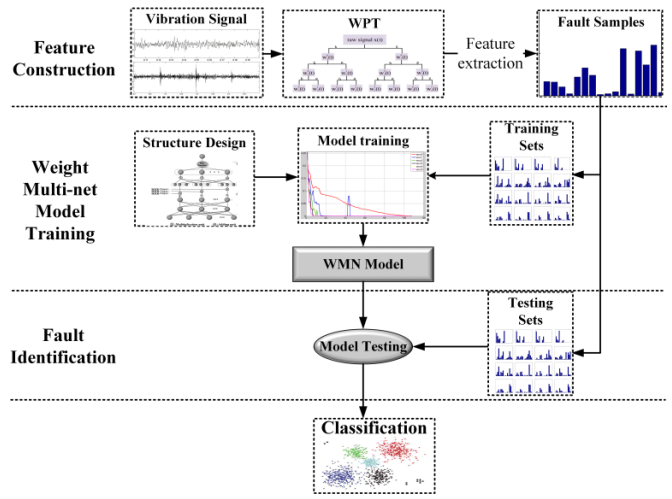

Fig. 6. The process of bearing fault identification via multi-weight fusion

IV. EXPERIMENT VERIFICATION

A. Data acquisition

To evaluate the performance of the proposed architecture for fault identification under unknown speed conditions, a series of experiments data were studied in this study. The experimental data with multiple bearing faults are provided by Case Western Reserve University Bearing Data Center [35]. The structure of the bearing fault testing bench is shown in Fig 7, which includes electric motor, torque sensor and power tester three parts. The fault bearings are the support bearing of the motor shaft on the fan end and driven end. The bearing types were SKF6205 and SKF6203. Three vibration acceleration sensors were respectively mounted on the base, fan end and drive end. And the bearing fault was formed by electrical discharge machining.

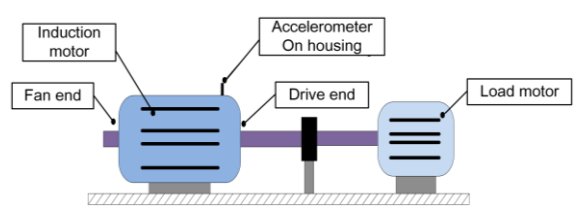

Fig. 7. Bearing fault experiment system

In this study, the vibration signal of the driven end was used as the experimental data. Normal and three bearing fault types (outer race defect, inner race defect and ball defect) were chosen, where the damage sizes of the each fault type included 0.007, 0.014 and 0.021 inches respectively. The vibration was

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measured with the sampling frequency of 12 kHz under four different speed conditions (1725/1750/1772/1796 rpm). Therefore, there is vibration samples of ten health conditions (normal and nine damage faults) under each speed, and these data was further used to validate the performance of the proposed method. The details of experimental information are shown in Table 1. For each health conditions under different speed, 100 samples (60 for training and 40 for testing) were built with the sample length of 2400 and coincidence degree of 50% between two signal segments**.**

TABLE 1 Information of bearing fault experiment

Bearing type	Fault type	Fault size (*inch*)	Speed (*rpm*)	Sampling frequency (kHz)
SKF 6203	Normal	0	1725/1750 /1772/1796	12
	Inner race defect	0.007/0.014/ 0.021		
	Outer race defect	0.007/0.014/ 0.021		
	Ball defect	0.007/0.014/ 0.021		

TABLE 2 The organization of training and testing date sets

Data sets	Speed conditions (rpm)	Number of training/ testing samples
A	1725	600/400
B	1750	600/400
C	1772	0/1000
D	1796	600/400

As introduced in Section II, a 4-level WPT (Daubechies44 wavelet [21]) was used and the corresponding reconstructed coefficients were further employed to construct the sixteen features of coefficients energy ratio. The details of the training and testing sets are shown in Table 2. It can be seen that four datasets (A, B, C and D) include ten bearing health conditions under four different speeds (1725, 1750, 1772 and 1796 rpm, are built, where 100 samples are constructed for each conditions. This indicates that the total number of the dataset is 1000. Specifically, for datasets A, B and D, 600 samples are chosen for training process and the rest ones are used for testing process. And these three datasets under different speeds were used to verify the effectiveness of the proposed WMN model in recognition effect of training information. Furthermore, dataset C under an unknown speed is directly used to validate the fault identification performance of the network for unknown speed conditions.

B. Multiple Feature Leaning under Different Speeds

As introduced in previous section, it can be known that three units, including identification unit, weight unit and fusion unit, play important roles in the proposed WMN. Therefore, to illustrate the effectiveness of this special architecture, these units have been verified in this section.

To analysis the learning effect of identification unit, the samples from datasets A, B and D are selected as training sets, thus as plotted in Fig. 5 for fault identification unit, three sub-nets (V_A_Net, V_B_Net, V_D_Net) are respectively built via the conventional BP-net. The corresponding feature clustering results of these three sub-nets can be obtained and shown in Fig. 8, where the first three features extracted by PCA are drawn here. It can be seen that the clustering boundaries of each fault type are obvious, which testifies that each sub-net in identification unit shows a high recognition effect for the samples under the same speed conditions. It means that each sub-net shows a good recognition effect for the same speed information that has been trained.

Meanwhile, for a better illustration of the property of the proposed WMN, two other neural networks are also structured and used to analysis these different sample sets, including SNN (single neural network: a single there layer ANN, with the number of neural neurons in three layers are 16, 24 and 10 respectively) and FNN (fusion neural network: consists of three SNN and a weight fusion layer, and the multiple results of identification unit are fused with equal weights, where these nets have the same parameters with SNN). We can find that SNN is the basic sub-net as compared the multiple sub-nets in WMN while FNN only has an identification unit without a dynamic weight as compared to the ones in WMN (the number of neural neurons in three layers of the weight unit in WMN are 16, 32 and 3 respectively). Both of SNN and FNN have no treatment strategy for the speed information representation.

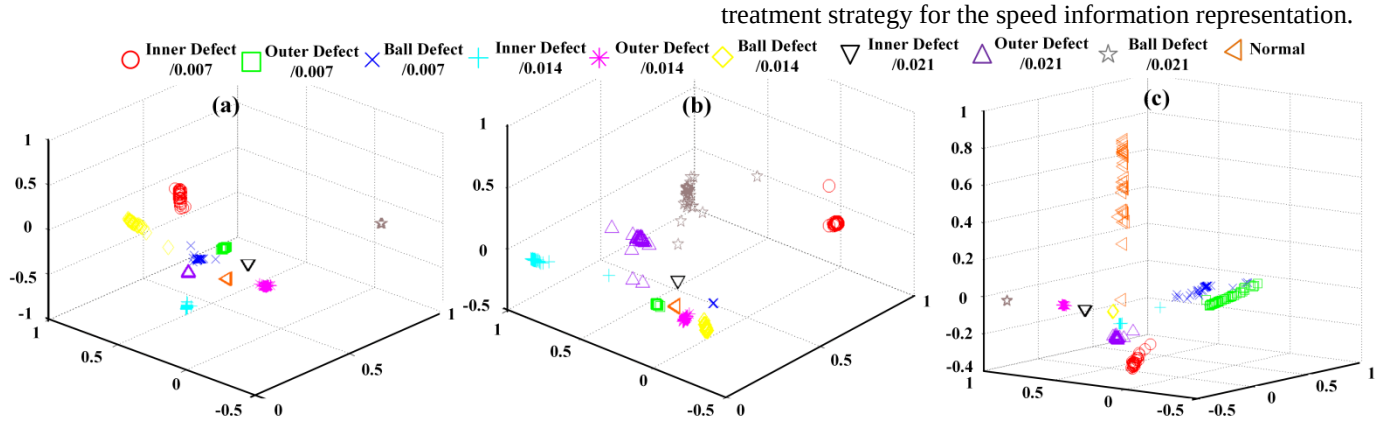

Fig. 8. Feature clustering plots for identification unit with ten different bearing fault types under different speeds (a) Set A, (b) Set B, (c) Set D

Firstly, SNN is trained by training samples of dataset D and tested by the testing sample of four datasets (A/B/C/D), and the study rate is adjusted by Eq. (19). The accuracies for those four datasets are shown in Table 3. Because the datasets A/B/C under different speed conditions are not included in training set, we can find that SNN only has a good classification precision for Set D while show a low classification accuracy for other unknown datasets. Then, another combined dataset A-B-D is

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together used as the training set of SNN, and the mentioned four datasets A/B/C/D are used for testing. From Table 3, it can be also found that the accuracies of datasets C/D are worsened while the accuracy of dataset B is also dissatisfied. This means that the conventional SNN has turned to fail for these complex fault types identification under different speed conditions.

TABLE 3 Classification performance of SNN

Training Sets	Testing Sets	Identification accuracy (%)	
D	Set A	85.50	87.75
	Set B	84.33	
	Set C	82.67	
	Set D	98.50	
A-B-D	Set A	100	86
	Set B	90.50	
	Set C	75.50	
	Set D	70	

TABLE 4 Classification performance of FNN

Training Sets	Testing Sets	Identification accuracy (%)				
		V_A_Net	V_B_Net	V_D_Net	FNN	
A-B-D	Set A	99.9	78.1	64.8	89.8	92.8
	Set B	89.6	99.7	81.8	99.7	
	Set C	80.6	95.0	89.6	95.0	
	Set D	87.8	86.7	87.2	86.7	

TABLE 5 Classification performance of WMN

Training Sets	Testing Sets	Identification accuracy (%)				
		V_A_Net	V_B_Net	V_D_Net	WMN	
A-B-D	Set A	99.9	78.1	64.8	99.8	98.8
	Set B	89.6	99.7	81.8	100	
	Set C	80.6	95.0	89.6	97.0	
	Set D	87.8	86.7	87.2	98.3	

TABLE 6 The weight distribution of different sub-nets

	$w1$(unit A)	$w2$(unit B)	$w3$(unit D)
Sample1 (A)	**0.614**	0.294	0.092
Sample2 (B)	0.279	**0.468**	0.253
Sample3 (C)	**0.301**	**0.358**	**0.341**
Sample4 (D)	0.0764	0.091	**0.832**

Secondly, FNN is used to deal with these same issues, and the results are shown in Table 4. The result shows that the accuracies of different sub-nets to the corresponding sets are gratifying, but the fusion results are not good. It seems even worse than the ones of SNN. In this manner, we will find that each sub-net can achieve a good recognition effect for its corresponding samples under same speed condition while the net will turn to fail for other condition. This means that a correct weight distribution is very important for fusion neural network.

Finally, according to the analysis of SNN and FNN, the proposed WMN is also tested for those same issues, where the weight unit is trained by dataset A-B-D and the model is tested by dataset A/B/C/D. As listed in Table 5, the proposed method not only has a high recognition accuracy for datasets A/B/D (which has been trained in the model), but also has a good performance for dataset C (which has not included in training sets). Meanwhile, as compared to the results of SNN and FNN, the proposed WMN shows the best classification performance with a high accuracy. This further indicates that the weight unit in WNN has powerful function in the problem of fault identification under complex working conditions.

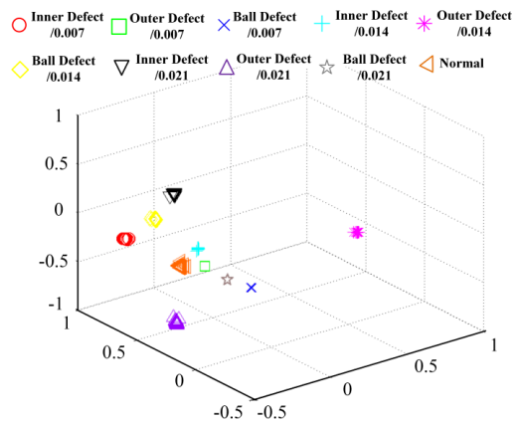

Fig. 9 Feature clustering plots based on WMN architecture for an unknown dataset C.

For a better illustration of the dynamic weight distribution, the samples under four different speed conditions are analyzed by the trained WMN model. The weight distribution provided by weight unit to each sub-net under corresponding to different speed conditions are shown in Table 6. A logical result is that the weight unit distributes a bigger weight value for the sub-net, whose training dataset has the same speed condition with the testing sample. For an example, the unit A is trained by dataset A, and the sample1 also belongs to dataset A. The weight value distributed to sub-net V_A_Net (0.614) is the biggest one compared to the ones of other two sub-nets (0.294 and 0.092). Nevertheless, for the sample3, because dataset C was not trained in the identification unit, the weight values distributed to three sub-nets are more uniform. This phenomenon is same as the explanation before. As a result, Fig. 9 shows the scatter clustering of WNN for the unknown testing dataset C. Benefit from the dynamic weight distribution calculated from weight unit, although the testing sample lies in an unknown speed conditions, the WNN model can accurately classifies these ten fault types with speed information ignored.

Furthermore, figure 10 illustrates the feature transformation process in two different networks, including traditional BP and the proposed WMN method, where their input features, middle high dimension features and output features are described by histogram statistical analysis. As shown in Fig. 10, the same row has different speed conditions while the same line has different fault types. From these sub-figs, it can be found that there are great difference in the input features of fault 2 and fault 3 under different speeds, which means that speed has serious influence to the features of this two fault types. For the output features of traditional BP method, it can be easily found that the differences between features under different speeds still exist. This will bring a miss understanding for the fault types, where low accuracy is achieved for fault identification under multiple speed conditions. However, due to the identification unit and weight unit in the proposed architecture, the output features for same fault types under different speeds show a similar distribution. This further indicates that the proposed WMN model discovers the implicit identical characteristics of same fault signals under different speeds and has high

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identification accuracy for sample under unknown speed condition.

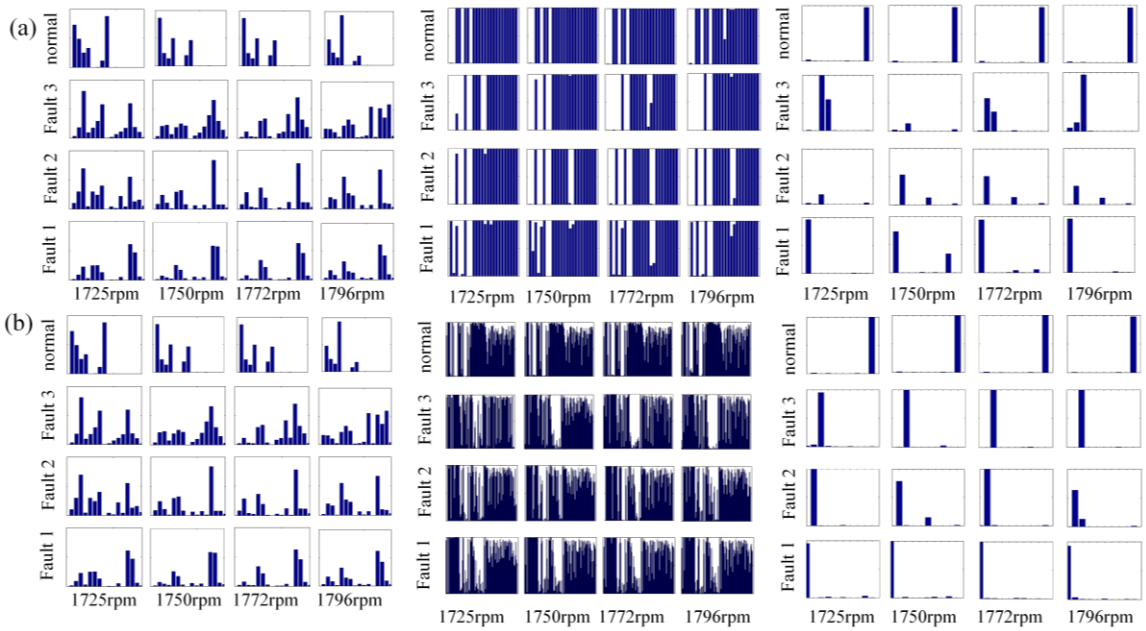

Fig. 10. The feature transition process for two different network architectures (a) traditional BP method and (b) proposed WMN method.

C. Classification Comparison

In order to demonstrate the superiority of the proposed WMN in identification of samples under unknown speed condition, the result analysis for the mentioned issues have been also compared with other three traditional classification methods, including K-means, principle component analysis (PCA) and typical BP method in this study. Specially, K-means method was used for the original features to distinguish the fault type. PCA method is applied to reduce the feature dimension with the principle components remained before K-means classifier. The traditional BP method is a 3-layer network with the layer number of 16, 36 and 10, respectively. In this manner, datasets A, B and D are first combined together to train the model for those models and the other four datasets A/B/C/D are further tested to verify the performance. The feature cluster distribution of those four methods for ten fault types under different speeds are displayed in Fig. 11. It can be found that the scatter plots based on K-means have serious fuzziness between classes, and only two types are separated obviously. For the results based on PCA method, due to the limitation of ability in feature extraction, there exists a phenomenon that not only more classes are distinguished but also the samples with same labels in fuzziness area are more convergent. And BP method can decrease more fuzziness between different labels in a certain degree, but there also exist small overlapping area between fault 2, 5, 6 and 8, which would bring difficulties for these fault types identification. Different from those features clustering, the feature based on the proposed WMN method has the best clustering effect, where those ten fault types under four different speeds are correctly distinguished and their boundaries between different types are clearly distributed.

To further quantitatively analyze clustering performance of the proposed WNN method, the accuracies of the four testing datasets based on four different methods are listed in Table 7. Here, datasets C and B are respectively used to imitate new datasets under unknown speed condition, which has been not included in the training samples. It can be seen that, for K-means, PCA, BP network and the proposed WMN, the average accuracies for those four dataset are incremental. The WNN method has the best performance. Additionally, once the dataset is not included in the training sets, the identification accuracy of this set based on K-means and PCA method are much lower than other datasets. Meanwhile, the total accuracies for this complex fault type under different speeds are also very terrible. Nevertheless, for the ones based on BP net, the accuracy of the selected dataset under an unknown speed shows a much lower accuracy than the datasets which are used as the training sets. This limitation has further weakened the ability of BP net in the identification for sample under uncertain speed working condition. Motivated by this issue, the proposed WMN synthesizes three special nets with the speed information mined as introduced in Section III. In this manner, even an input sample is not included in the training set, the proposed WMN has good recognition accuracy for all testing sets as illustrated in Table 7. The proposed method shows the highest accuracy and a relatively stable recognition effect for the samples under complex speed working condition. This further indicates that the proposed method can effectively remove the influence of speed conditions to identification result with weight unit.

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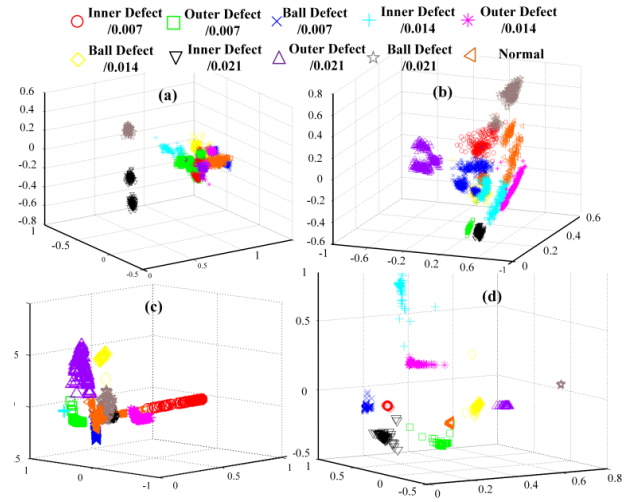

Fig. 11. Feature clustering of different methods (a) K-means (b) PCA (c) ANN method and (d) Proposed WMN method.

TABLE 7 Classification performance by different methods

Training Sets	Recognition Accuracy（%）				
	Testing Sets	K-means	PCA	BP	WNN
A-B-D	**Average**	**66.98**	**75.65**	**96.96**	**98.77**
	A	70.00	65.00	97.50	99.75
	B	69.90	79.80	99.17	100.00
	C	70.00	79.80	91.17	97.00
	D	58.00	78.00	100	98.25
A-C-D	**Average**	**71.50**	**80.38**	**86.00**	**99.31**
	A	67.20	63.40	100.00	99.50
	B	70.10	81.80	90.50	99.50
	C	78.70	89.90	75.70	100
	D	70.00	86.40	78.00	98.25

D. Discussion

(1) Compared with traditional networks, WMN consists of identification unit, a weight unit and a fusion unit, where the analysis has verified that these multiple sub-nets in identification unit have high classification accuracy for the testing samples under their corresponding speed conditions, and the weight unit can distribute correct weights to these sub-nets according to the distribution of the input features. Furthermore, it can be also found that the influence of operation conditions is crucial for fault diagnosis, and much more working conditions, such as speed, load and temperature, should be considered in the process of robust but sensitive feature construction. In this manner, the Network construction is the crucial work in the further.

(2) It has been verified that the proposed method has an ideal fault recognition capability under complex speed conditions by the experimental results. Since the WMN model adopts the idea of working condition interpolation and is trained by the samples under multiple typical speeds, the adjustment of the fusion weights are realized according to the information of working condition in the input sample, and the influence of working condition to fault identification results has considered. However, according to the principle of working condition interpolation, it can be foreseen the WMN model also has to run into the effect of working condition boundary. For a better illustration, the fault identification results of the WMN model under the other combinations, including training sets A-B-C and training sets B-C-D, are also tested here in the same condition, the fault recognition accuracy under the boundary conditions (dataset D and dataset A) are 94.33% and 83.67% respectively. It can be seen that, once the training sets did not include the boundary conditions, the corresponding fault identification accuracies of the WMN model under these boundary conditions are much lower than the results in Table 7. This further indicates that the proposed WMN model in an idea of working condition interpolation for an unknown dataset shows an effect of working condition boundary and the recognition ability will be weaken. Focusing this shorting, our following research emphasis is to remove the effect of working condition boundary for this weight architecture.

(3) In the process of WMN construction, the vital important part is to choose the typical speed conditions, which is based the distribution law analysis of speed characteristic. Due to only four speed conditions existed in database of Case Western Reserve University Bearing Data Center, this section was simplified to choose the boundary speeds and any one intermediate speed as the speed nodes. But, once the working conditions were more complex, this issue should be deserved special attention. In this manner, for WMN architecture, there still exist some issues to be studied such as the selection of typical speed nodes under much more working conditions and dynamic weight adjustment rule. This will be studied in the further work for a sound and better performance in the application of fault classification under complex working conditions.

(4) In this study, a three-layer BP network is used as the basic architecture of the proposed WMN. It can be seen that the proposed method focuses on the design of the identification unit, a weight unit and a fusion unit, where the speed information can be well represented in the nets for a better understanding of the machine health. Meanwhile, we can also found that a much better effect can be achieved as compared the typical architecture of BP. In these manners, it can be foreseen that many other deep learning architecture can be also employed here to accomplish a sound and intelligent fault diagnosis with regardless of the speed, load, temperature and many other working condition.

V. Conclusions

This study proposed a new weight multi-net for bearing fault identification under unknown speed conditions. The principle contribution is to conduct different network units to extract the fault information and the influence law of speed to fault features. By combining the analysis result of different information in a fusion process, the multiple units via dynamic weight fusion are synthesized in a speed-insensitive network architecture. Firstly, with the energy distribution of wavelet packet segmentation coefficients input, fault identification unit is built where the fault information for different typical speed conditions are remained in the corresponding sub-nets. Then, the weight unit is later structured to contribute appropriate weights for those multiple results of the identification unit. The final identification result is obtained by fusing the identified

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results with the weights in fusion unit. Compared with other three traditional classification methods, it can be seen that the proposed WMN not only has a good identification performance for multiple speed conditions, but also shows a strong ability to unknown information representation for samples under uncertain speed condition.

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